PROBLEM 22

Particle on a Curve

Original statement · Problems.pdf, p. 10

1D Curve Constraint

A heavy point particle P of mass m is constrained to move without friction along the curve:

$$y = a\left(1 - \cos\frac{x}{a}\right) \tag{4}$$

in the range $-\pi a \leq x \leq \pi a$, where a is a positive constant. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

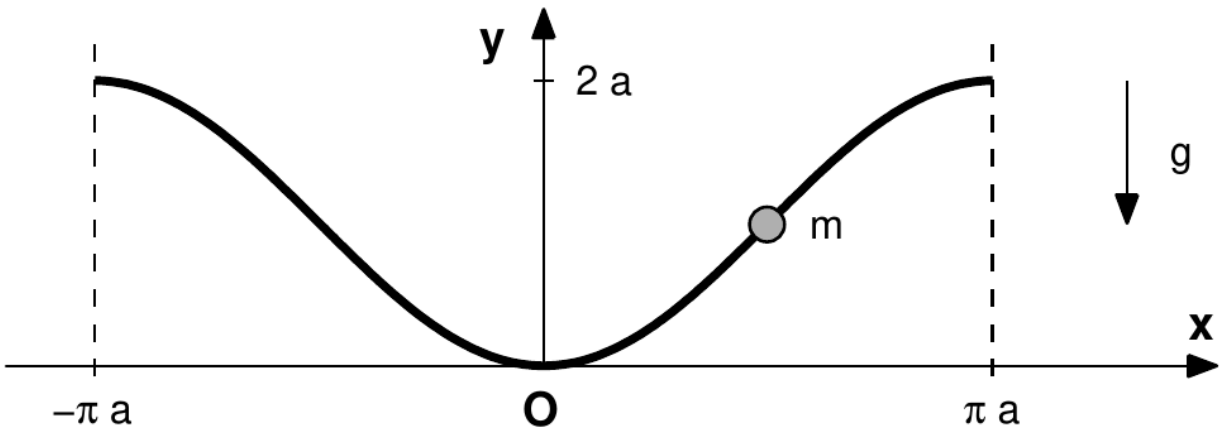


Figure: Heavy particle m on the curve y = a(1 - cos(x/a)) between x = -pi a and x = pi a, gravity g downward (Problems.pdf, p. 10).

(a) Reduce the problem to integrals.

(b) If $v_0 = \sqrt{ga}$, determine the minimum and maximum value of x along the motion.

(c) What should be the minimum value of v_0 in order to reach $x = \pi a$?

Let's assume now that the curve containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega \mathbf{j}$.

(d) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Determine the inertia force that acts on the particle in this moving reference frame as a function of m, Ω, x_1 and $\dot{x}_1$, where x_1 represents the position coordinate of P along OX_1.

(e) Reduce the problem to quadratures and determine the value of the reaction force.

(f) Find out a trascendental [sic] equation whose solutions are the equilibrium positions of the point particle in $OX_1Y_1Z_1$.

(g) If Ω were a function of time, would it be possible to use the energy equation in the same way as before? Justify your answer.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **Physical System:** A particle of mass m is constrained to move without friction along the smooth planar curve:

$$y = a\left(1 - \cos\frac{x}{a}\right), \qquad x \in [-\pi a, \pi a]$$

in a vertical plane under uniform downward gravity $\mathbf{g} = -g\mathbf{j}$.

- **Degrees of freedom:** 1 degree of freedom, parametrized by Cartesian abscissa $x(t)$.
- **Initial State ($t = 0$):** $x(0) = 0 \implies y(0) = a(1 - \cos 0) = 0$, with initial velocity $\mathbf{v}_0 = v_0\mathbf{i}$ ($v_0 > 0$).

- **Parameters:** $m > 0$ [kg], $a > 0$ [m], $g > 0$ [m/s^2], $v_0 > 0$ [m/s].

PHASE 2

Reference Frames & Vector Bases

- **Inertial Frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$.
- **Curve Differential Elements:**

$$\frac{dy}{dx} = \frac{d}{dx}\left[a\left(1 - \cos\frac{x}{a}\right)\right] = \sin\left(\frac{x}{a}\right)$$

Differential arc-length:

$$ds^2 = dx^2 + dy^2 = dx^2 + \sin^2\left(\frac{x}{a}\right)dx^2 = \left[1 + \sin^2\left(\frac{x}{a}\right)\right]dx^2$$

$$ds = \sqrt{1 + \sin^2\left(\frac{x}{a}\right)}\,dx$$

- **Velocity:**

$$v = \frac{ds}{dt} = \sqrt{1 + \sin^2\left(\frac{x}{a}\right)}\,\dot{x}$$

- **Rotating frame (parts d to g).** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$ turns with the curve about $\mathbf{j}$ through the angle Ωt, $[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}$, with $\det = \cos^2\Omega t + \sin^2\Omega t = 1$ and $[{}_0R_1][{}_0R_1]^T = I$. Its origin is fixed and $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$.

PHASE 3

Derivation

3.1 Kinematics

The particle moves on $y = a\,(1 - \cos(x/a))$, so $\dot{y} = \sin(x/a)\,\dot{x}$ (chain rule) and its speed is $v = \sqrt{1 + \sin^2(x/a)}\,\dot{x}$ (Phase 2); the unit tangent is $\mathbf{e}_t = [\mathbf{i} + \sin(x/a)\,\mathbf{j}]\,/\sqrt{1 + \sin^2(x/a)}$.

3.2 Forces

The weight $-mg\,\mathbf{j}$ (conservative, $V = mgy$) and the reaction $\mathbf{N}$ of the smooth curve, perpendicular to $\mathbf{e}_t$, so $\mathbf{N} \cdot \mathbf{v} = 0$. In part (d) the inertia forces of the rotating frame are added.

3.3 Equations of Motion

Projecting Newton's second law on $\mathbf{e}_t$ removes $\mathbf{N}$; the energy integral used below is the first integral of that tangential equation (work-energy theorem). Written for $x(t)$: differentiating $\tfrac{1}{2}m(1 + \sin^2(x/a))\dot{x}^2 + mga(1 - \cos(x/a)) = \text{const}$ with the product rule and the chain rule ($\frac{d}{dt}\sin^2\frac{x}{a} = \frac{2}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}$, $\frac{d}{dt}\cos\frac{x}{a} = -\frac{1}{a}\sin\frac{x}{a}\,\dot{x}$) gives $m\left(1 + \sin^2\frac{x}{a}\right)\dot{x}\ddot{x} + \frac{m}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}^3 + mg\sin\frac{x}{a}\,\dot{x} = 0$, and dividing by $m\dot{x}$ ($\dot{x} \neq 0$):

$$\left(1 + \sin^2\frac{x}{a}\right)\ddot{x} + \frac{1}{a}\sin\frac{x}{a}\cos\frac{x}{a}\,\dot{x}^2 + g\sin\frac{x}{a} = 0$$

In the rotating frame (parts d to g) the tangential projection of the centrifugal force $m\Omega^2 x_1\mathbf{i}_1$ adds $\Omega^2 x_1$ to the right-hand side:

$$\left(1 + \sin^2\frac{x_1}{a}\right)\ddot{x}_1 + \frac{1}{a}\sin\frac{x_1}{a}\cos\frac{x_1}{a}\,\dot{x}_1^2 + g\sin\frac{x_1}{a} = \Omega^2 x_1$$

3.4 Integration and Answers

Part (a): Reduction of Motion to Integrals

Pedagogical Justification: The frictionless constraint does zero work ($W_N = 0$). Gravity is conservative with potential energy $V(y) = mgy = mga(1 - \cos(x/a))$. Total mechanical energy is constant.

$$E = \frac{1}{2}mv^2 + mgy = \frac{1}{2}mv_0^2 + mg(0) = \frac{1}{2}mv_0^2$$

$$\frac{1}{2}m\left[1 + \sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2 + mga\left(1 - \cos\frac{x}{a}\right) = \frac{1}{2}mv_0^2$$

Multiply by $2/m$:

$$\left[1 + \sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2 = v_0^2 - 2ga\left(1 - \cos\frac{x}{a}\right)$$

Isolate $\dot{x} = \frac{dx}{dt}$:

$$\frac{dx}{dt} = \pm\sqrt{\frac{v_0^2 - 2ga(1 - \cos(x/a))}{1 + \sin^2(x/a)}}$$

Separation of variables yields the quadrature:

$$\int_0^{x(t)} \sqrt{\frac{1 + \sin^2(x'/a)}{v_0^2 - 2ga(1 - \cos(x'/a))}}\, dx' = \pm t$$

Part (b): Turning Points for $v_0 = \sqrt{ga}$

> **Pedagogical Justification:** Turning points occur where the velocity vanishes ($v = 0 \iff \dot{x} = 0$).

$$v_0^2 - 2ga\left(1 - \cos\frac{x}{a}\right) = 0$$

Substitute $v_0^2 = ga$:

$$ga - 2ga\left(1 - \cos\frac{x}{a}\right) = 0$$

Divide by ga:

$$1 - 2\left(1 - \cos\frac{x}{a}\right) = 0 \implies 2\cos\frac{x}{a} = 1 \implies \cos\frac{x}{a} = \frac{1}{2}$$

In the domain $x/a \in [-\pi, \pi]$:

$$\frac{x}{a} = \pm\frac{\pi}{3} \implies x_{min} = -\frac{\pi}{3}a, \qquad x_{max} = +\frac{\pi}{3}a$$

Part (c): Minimum Velocity v_0 to Reach $x = \pi a$

> **Pedagogical Justification:** At the crest $x = \pi a$, the height is $y(\pi a) = a(1 - \cos\pi) = a(1 - (-1)) = 2a$. For the particle to reach this crest, its kinetic energy at the peak must be non-negative ($T(\pi a) \geq 0$).

$$E \geq V(\pi a) \implies \frac{1}{2}mv_0^2 \geq mg(2a) = 2mga$$

$$v_0^2 \geq 4ga \implies v_{0,min} = \sqrt{4ga} = 2\sqrt{ga}$$

At exactly $v_0 = 2\sqrt{ga}$ the crest is a maximum of V (zero force there), so it is reached only asymptotically ($t \to \infty$); for $v_0 > 2\sqrt{ga}$ it is reached in finite time. $2\sqrt{ga}$ is therefore the infimum of the admissible values.

Part (d): Moving Reference Frame with Rotation $\boldsymbol{\omega} = \Omega\mathbf{j}$

- Frame $\mathcal{F}_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$ rotating with $\boldsymbol{\Omega} = \Omega\mathbf{j}$.
- Particle position: $\mathbf{r}' = x_1\mathbf{i}_1 + y(x_1)\mathbf{j}$.
- Relative velocity: $\mathbf{v}' = \dot{x}_1\mathbf{i}_1 + \sin(x_1/a)\dot{x}_1\mathbf{j}$.
- Inertia forces in $\mathcal{F}_1$ (the origin is fixed and Ω is constant, so $\mathbf{F}_A = \mathbf{F}_E = \mathbf{0}$; the total inertia force is $\mathbf{F}_i = m\Omega^2 x_1\,\mathbf{i}_1 + 2m\Omega\dot{x}_1\,\mathbf{k}_1$, from the two terms below):

1. Centrifugal Force:

$$\boldsymbol{\Omega}\times\mathbf{r}' = \Omega\,\mathbf{j}\times(x_1\mathbf{i}_1 + y\,\mathbf{j}) = -\Omega x_1\,\mathbf{k}_1, \qquad \mathbf{F}_{cf} = -m\boldsymbol{\Omega}\times(\boldsymbol{\Omega}\times\mathbf{r}') = -m\,\Omega\,\mathbf{j}\times(-\Omega x_1\mathbf{k}_1) = m\Omega^2 x_1\mathbf{i}_1$$

2. Coriolis Force:

$$\mathbf{F}_{Cor} = -2m\boldsymbol{\Omega}\times\mathbf{v}' = -2m(\Omega\mathbf{j})\times(\dot{x}_1\mathbf{i}_1 + \dot{y}_1\mathbf{j}) = 2m\Omega\dot{x}_1\,\mathbf{k}_1$$

Part (e): Quadratures and Reaction Force in the Moving Frame

- Effective potential in $\mathcal{F}_1$:

$$V_{eff}(x_1) = mgy(x_1) - \frac{1}{2}m\Omega^2 x_1^2 = mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2$$

- Energy conservation in $\mathcal{F}_1$ (the curve is fixed and smooth in $\mathcal{F}_1$, so $\mathbf{N}\cdot\mathbf{v}' = 0$; $\mathbf{F}_{Cor}\cdot\mathbf{v}' = 0$ always; gravity is conservative and $V_{cf} = -\frac{1}{2}m\Omega^2 x_1^2$ does not depend on time because Ω is constant):

$$E' = \frac{1}{2}m\left[1 + \sin^2(x_1/a)\right]\dot{x}_1^2 + mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2 = \text{constant}$$

- **Initial value.** At $x_1 = 0$ the position is $\mathbf{0}$, so $\boldsymbol{\Omega}\times\mathbf{r}' = \mathbf{0}$ and the relative speed equals v_0: $E' = \frac{1}{2}mv_0^2$. Solving for $\dot{x}_1$ and separating variables (Barrow's rule from $x_1(0) = 0$):

$$\dot{x}_1^2 = \frac{v_0^2 - 2ga\left(1-\cos\frac{x_1}{a}\right) + \Omega^2 x_1^2}{1+\sin^2(x_1/a)} \implies \pm t = \int_0^{x_1} \sqrt{\frac{1+\sin^2(\xi/a)}{v_0^2 - 2ga\left(1-\cos(\xi/a)\right) + \Omega^2\xi^2}}\, d\xi$$

- **Reaction.** $\mathbf{N} = N_n\,\mathbf{n} + N_z\,\mathbf{k}_1$ with the in-plane unit normal $\mathbf{n} = \left(-\sin\frac{x_1}{a}\,\mathbf{i}_1 + \mathbf{j}\right)/\sqrt{1+\sin^2(x_1/a)}$ (perpendicular to the tangent $\mathbf{i}_1 + \sin\frac{x_1}{a}\,\mathbf{j}$).

Along the binormal $\mathbf{k}_1$: $N_z = -F_{Cor,z} = -2m\Omega\dot{x}_1$.

Along $\mathbf{n}$: the relative acceleration is $\mathbf{a}_1 = \ddot{x}_1\left(\mathbf{i}_1 + \sin\frac{x_1}{a}\,\mathbf{j}\right) + \frac{\dot{x}_1^2}{a}\cos\frac{x_1}{a}\,\mathbf{j}$ (product and chain rules), so $m\,\mathbf{a}_1\cdot\mathbf{n} = \dfrac{m\dot{x}_1^2\cos(x_1/a)}{a\sqrt{1+\sin^2(x_1/a)}}$; the weight gives $-mg/\sqrt{1+\sin^2(x_1/a)}$ and the centrifugal force $-m\Omega^2 x_1\sin(x_1/a)/\sqrt{1+\sin^2(x_1/a)}$. Hence

$$N_n = \frac{m}{\sqrt{1+\sin^2(x_1/a)}}\left[g + \Omega^2 x_1\sin\frac{x_1}{a} + \frac{\dot{x}_1^2}{a}\cos\frac{x_1}{a}\right]$$

with $\dot{x}_1^2$ from the energy integral above.

Part (f): Transcendental Equilibrium Equation

Equilibrium occurs where $\frac{dV_{eff}}{dx_1} = 0$:

$$\frac{dV_{eff}}{dx_1} = mg\sin\left(\frac{x_1}{a}\right) - m\Omega^2 x_1 = 0 \implies g\sin\left(\frac{x_1}{a}\right) - \Omega^2 x_1 = 0$$

Part (g): Case $\Omega = \Omega(t)$

If Ω varies with time, an Euler force $\mathbf{F}_{Euler} = -m\dot{\Omega}\,\mathbf{j}\times\mathbf{r}' = m\dot{\Omega}x_1\,\mathbf{k}_1$ appears; it is normal to the plane of the curve, so $\mathbf{F}_{Euler}\cdot\mathbf{v}' = 0$ and it does no work in $\mathcal{F}_1$ (it is balanced by N_z). The energy is lost for another reason: the centrifugal potential depends explicitly on time, $\frac{\partial V_{cf}}{\partial t} = -m\Omega\dot{\Omega}x_1^2 \neq 0$, so $\frac{dE'}{dt} = -m\Omega\dot{\Omega}x_1^2 \neq 0$. Thus the mechanical energy in $\mathcal{F}_1$ is **not** conserved and the energy equation cannot be used in the same way.

PHASE 4

Interpretation & Checks

- The potential well has width $2\pi a$ and barrier height $2a$ (potential energy $2mga$). Rotation creates a centrifugal destabilizing potential $-\frac{1}{2}m\Omega^2 x_1^2$ so that the origin loses stability for $\Omega^2 > g/a$. Off-centre equilibria satisfy $\sin u = \lambda u$ with $u = x_1/a$ and $\lambda = \Omega^2 a/g$; since $\sin u/u$ decreases monotonically from 1 to 0 on $(0, \pi]$, they exist (one pair $\pm u^*$ with $0 < u^* < \pi$) only for $\lambda < 1$, i.e. $\Omega^2 < g/a$, and there they are unstable ($V''_{eff} = (mg/a)(\cos u^* - \lambda) < 0$).
- **Limit $\Omega \to 0$:** $V_{eff} \to mga\,(1 - \cos(x_1/a))$, $N_z \to 0$, the quadrature of (e) reduces to that of (a) and the equilibrium equation $g\sin(x_1/a) = 0$ leaves only $x_1 = 0$ inside $(-\pi a, \pi a)$.
- **Numeric check:** a scan of $\sin u = \lambda u$ on $(0, \pi]$ finds one root for $\lambda = 0.5$ ($u^* = 1.8955$) and for 0.9, and none for $\lambda = 1, 1.1, 2$.
- **Dimensions:** $[g/a] = [\Omega^2] = \mathrm{s}^{-2}$, so $[\lambda] = [\Omega^2 a/g] = 1$; every term of E' is in J ($[mga] = [m\Omega^2 x_1^2] = [mv_0^2] = \mathrm{kg\,m^2 s^{-2}}$); $[N_z] = [m\Omega\dot{x}_1] = \mathrm{N}$; $[g\sin(x/a)] = [\Omega^2 x_1] = \mathrm{m\,s^{-2}}$.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 22 in its solution keys, so the results are verified by independent derivation (including the numeric scan of the equilibrium equation in Phase 4).

Source: Problems.pdf